

# allocsys

Backend Developer — APIs, Infra & AI-Agent Tooling

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## SUMMARY

Backend-focused developer who builds the infrastructure layer AI agents actually run on — auth, rate limiting, multi-service integrations — rather than just wiring prompts together. Comfortable owning a project end-to-end: designing the API surface, hardening it (timing-safe auth, IP allowlisting, proxy-aware rate limiting), and documenting the reasoning so someone else can maintain it. Recent focus: Model Context Protocol (MCP) servers and agentic workflow automation (n8n + LangChain), plus root-causing bugs in real-world open-source infrastructure.

## PROJECTS

### madmcp

*Node.js / Express · MCP server*

- Built and deployed a single MCP server exposing ~50 tools across 5 backends (GitHub, Cloudflare D1/KV/R2/Workers, Notion, Mem0, generic fetch) to any MCP-compatible AI client.
- Designed the auth layer: timing-safe shared-key comparison, IP allowlisting scoped to Claude's published connector range, and per-route rate limiting, applied in the correct order so a leaked key alone can't reach the server.
- Handled real deployment edge cases (proxy-hop-aware client IP trust, raised body limits for large file pushes, graceful per-connector token failure) and shipped one-click deploy configs for Render and Vercel.

### Agentic Email Triage & Ops Automation

*n8n · LangChain · RAG*

- Designed an n8n workflow where an AI agent classifies inbound support email, grounds answers in a RAG-searched knowledge base, and drafts replies only for non-urgent cases.
- Built a safety boundary directly into the system prompt: escalation-worthy emails are flagged for a human and never auto-replied to — the boundary lives before generation, not as a filter after.
- Forced structured JSON output so downstream nodes (Sheets, CRM, LINE) route deterministically instead of parsing free text; tested against 6 representative scenarios.

### cf-manager

*Bash · Cloudflare API v4 · Termux*

- Built a self-contained bash script giving Cloudflare a full menu-driven admin panel (Workers deploy/rollback, KV/D1/R2/Hyperdrive) usable entirely from a phone terminal.
- Implemented OAuth2 + PKCE login with multi-account switching, one-tap rollback via local deploy backups, and a GitHub sync step that pushes a worker's source to a linked repo on deploy.

### domap

*Node.js · Playwright · Browserless.io*

- Built a headless-Chromium tool that walks a page's live DOM into a structured, selector-annotated tree, logs full network traffic, and replays recorded actions in a fresh session.
- Worked around Android's lack of native Playwright browser support by connecting to a remote Browserless.io session instead of a locally-launched binary.

## OPEN SOURCE CONTRIBUTIONS

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### drizzle-team/drizzle-orm

3 PRs · closed, not merged

- Fixed a node-postgres connection-pool leak on a rejected BEGIN, a TypeScript inference bug in createSchemaFactory, and added schema validation for 6 Gel-only column types across all four schema-integration packages.
- All three PRs were closed unmerged; the connection-pool fix's underlying issue remains open, with an equivalent fix from another contributor pending review.

### cloudflare/workers-sdk

4 PRs · 1 merged, 2 closed, 1 open

- Root-caused a Windows-only crash from non-ASCII paths (Latin-1 header violation) across 5 call sites — merged.
- Fixed a false infinite-redirect-loop deploy warning — closed, not merged.
- Fixed local Images binding transform parity (fit/gravity/background) — original PR closed after an accidental fork deletion mid-review; restored and continued in a follow-up PR, currently open and in review.

### urllib3/urllib3

PR #5121 · closed (superseded)

- Replaced a deprecated pyOpenSSL X.509 API call with the cryptography library equivalent, with new tests for both a present and missing commonName.

## SKILLS

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|--------------------|----------------------------------------------------------------------------------------|
| Languages          | JavaScript / TypeScript, Node.js, Python, SQL                                          |
| Backend & APIs     | Express, REST API design, OAuth2, Rate limiting, IP allowlisting, Webhooks             |
| AI / Agent Tooling | Model Context Protocol (MCP), LangChain, RAG / Vector DBs (Qdrant), n8n, Prompt design |
| Infrastructure     | Git, Cloudflare (D1, KV, R2, Workers), Render / Vercel, GitHub Actions CI              |